

• Homework 4

Using the same propulsion parameters used in lectures for medium-altitude nonlinear orbit control, and assuming J_2 as the only perturbation, apply the variable-thrust nonlinear orbit control to the following two cases:

(A) Initial orbit

$$p_0 = 18000 \text{ Km}, e_0 = 0.12, i_0 = 45 \text{ deg}, \Omega_0 = 30 \text{ deg}, \omega_0 = 0, \theta_{x0} = 0$$

Target orbit

$$p_d = 20000 \text{ Km}, e_d = 0, i_d = 50 \text{ deg}$$

(B) Initial orbit

$$p_0 = 8000 \text{ Km}, e_0 = 0.12, i_0 = 45 \text{ deg}, \Omega_0 = 30 \text{ deg}, \omega_0 = 0, \theta_{x0} = 0$$

Target orbit

$$p_d = 10000 \text{ Km}, e_d = 0, i_d = 50 \text{ deg}$$

For each case

- (i) portray the time histories of p, e, i, x_7 in $0 \leq t \leq 30$ days
- (ii) indicate the final values of p, e, i, x_7
- (iii) portray the time history of the Lyapunov function

Moreover

- (iii) justify the different behavior (for large times) between cases (A) and (B), including the plot for x_7 (mass ratio)

P.S. For the gains, use the same values employed in class for the nonlinear orbit control along medium-altitude orbits.